

Azure Master Data Engineer

6-Month Exhaustive Training & Certification Program

Advanced Architecture, Hands-on Labs, and MLOps

Abstract

This 24-week program is designed to transform you from a practitioner to a Master Data Engineer on Microsoft Azure. It progressively builds from core ADLS Gen2 data lakes to advanced Microsoft Purview governance, integrating daily theory, weekly service-specific mini-projects, architectural use case analyses, and massive monthly Capstone projects. A tracking timetable is included at the end.

Contents

1	Milestone 1: Core Data Infrastructure & Storage (Month 1)	2
2	Milestone 2: Data Warehousing & Analytics (Month 2)	3
3	Milestone 3: Batch Processing & ETL Orchestration (Month 3)	4
4	Milestone 4: Streaming & Real-Time Analytics (Month 4)	6
5	Milestone 5: Machine Learning Operationalization (Month 5)	7
6	Milestone 6: Security, Governance & Exam Prep (Month 6)	9
7	6-Month Progress Tracking Timetable	11

1 Milestone 1: Core Data Infrastructure & Storage (Month 1)

Objective: Master the foundational storage options, relational databases, and NoSQL.

Week 1: ADLS Gen2 & Storage Foundations

- **Monday:** Azure Storage Architecture, Blob Storage vs. Azure Data Lake Storage (ADLS) Gen2.
- **Tuesday:** Access Tiers (Hot, Cool, Cold, Archive), Lifecycle Management, and Soft Delete.
- **Wednesday:** Azure Data Box, AzCopy, and Azure Storage Explorer.
- **Thursday: Hands-on Project:** Deploy an ADLS Gen2 account via Azure CLI, configure a 30-day lifecycle transition to Archive tier, and set up a hierarchical namespace.
- **Friday: Use Case & Architecture:** Design a multi-region (Geo-Zone-Redundant), compliant disaster recovery storage architecture for a healthcare provider processing 10PB of X-ray images.

Week 2: Relational Databases (Azure SQL & PostgreSQL)

- **Monday:** Azure SQL Database: DTU vs. vCore models, Serverless, and Hyperscale.
- **Tuesday:** Azure Database for PostgreSQL: Flexible Server and High Availability (HA).
- **Wednesday:** Active Geo-Replication and Auto-failover groups.
- **Thursday: Hands-on Project:** Deploy an Azure SQL Database Serverless tier, create a geo-replica in another region, and simulate a regional failover.
- **Friday: Use Case & Architecture:** Architect a zero-downtime migration strategy from a monolithic on-premise SQL Server to Azure SQL Managed Instance.

Week 3: High-Scale NoSQL (Azure Cosmos DB)

- **Monday:** Cosmos DB core concepts: Request Units (RUs), Partitioning, and Global Distribution.
- **Tuesday:** Consistency Levels (Strong, Bounded Staleness, Session, Consistent Prefix, Eventual).
- **Wednesday:** APIs (Core SQL, MongoDB, Cassandra) and Cosmos DB Change Feed.
- **Thursday: Hands-on Project:** Create a Cosmos DB account (SQL API), design a single-container schema for a forum, write a Python script to insert items, and configure multi-region writes.
- **Friday: Use Case & Architecture:** Design an ingestion and serving layer for a gaming leaderboard supporting 1 million reads/writes per second globally with Session consistency.

Week 4: Hybrid Ingestion & Caching

- **Monday:** Azure Cache for Redis for database acceleration.
- **Tuesday:** Cosmos DB Integrated Cache.
- **Wednesday:** Azure API Management (APIM) for data ingestion.
- **Thursday: Hands-on Project:** Deploy an Azure Cache for Redis instance. Write a Python script to cache query results from an Azure SQL Database to reduce load.
- **Friday: Use Case & Architecture:** Architect a high-throughput, low-latency data API for a mobile application that experiences extreme traffic spikes during live sports events.

CAPSTONE PROJECT - Milestone 1: Build an E-Commerce Multi-Tier Data Backend. Deploy Azure SQL Database Serverless for user profiles/billing, Cosmos DB for active shopping carts and clickstream tracking, ADLS Gen2 for product images, and set up Cosmos DB Change Feed to push events to storage.

2 Milestone 2: Data Warehousing & Analytics (Month 2)

Objective: Deep dive into Azure Synapse Analytics, Lakehouses, and Purview.

Week 5: Azure Synapse Analytics (Dedicated SQL Pools)

- **Monday:** Synapse Architecture (formerly SQL DW), MPP architecture, and Data Warehouse Units (cDWUs).
- **Tuesday:** Distribution Types (Hash, Round-Robin, Replicate).
- **Wednesday:** Data ingestion: PolyBase vs. COPY statement.
- **Thursday: Hands-on Project:** Provision a Synapse Dedicated SQL Pool. Load 10 million rows of synthetic Parquet data from ADLS Gen2 using the COPY command with Managed Identity.
- **Friday: Use Case & Architecture:** Estimate the monthly cost and recommend an architecture (Dedicated vs. Serverless) for a BI workload heavily utilized only during end-of-month reporting.

Week 6: Synapse Optimization & Performance

- **Monday:** Indexing (Clustered Columnstore, Heap, Non-clustered).
- **Tuesday:** Partitioning strategies in Synapse Dedicated Pools.
- **Wednesday:** Workload Management (Workload Groups and Classifiers) and Materialized Views.
- **Thursday: Hands-on Project:** Create three identical tables in Synapse using different Distribution Types. Benchmark the execution time of a complex JOIN query on each table.
- **Friday: Use Case & Architecture:** Re-architect a poorly performing multi-petabyte analytics table that is causing CPU bottlenecks during complex daily dashboard aggregations.

Week 7: Data Lakehouse & Synapse Serverless SQL

- **Monday:** Synapse Serverless SQL Pools: Querying directly over ADLS Gen2 without provisioning compute.
- **Tuesday:** Logical Data Warehouse pattern (Views and External Tables).
- **Wednesday:** Delta Lake format concepts (ACID transactions on data lakes).
- **Thursday: Hands-on Project:** Write a Serverless SQL Pool query to read raw CSVs in ADLS Gen2, transform them on the fly, and save the output as Parquet (CETAS) back to the lake.
- **Friday: Use Case & Architecture:** Design a cost-effective "Lakehouse" architecture for a media company wanting to store 50PB of logs in ADLS Gen2 while querying them alongside core business data.

Week 8: Data Governance & Microsoft Purview

- **Monday:** Microsoft Purview: Centralized data map and cataloging.
- **Tuesday:** Data Lineage and automated classification of sensitive data.
- **Wednesday:** Azure Data Share for sharing data across Azure tenants.
- **Thursday: Hands-on Project:** Configure Microsoft Purview over an ADLS Gen2 bucket and Synapse workspace. Run a scan to automatically classify "PII_Email" columns and trace lineage.
- **Friday: Use Case & Architecture:** Architect a Data Clean Room for an ad agency to merge their CRM data with third-party advertiser data using Azure Data Share without exposing raw PII.

CAPSTONE PROJECT - Milestone 2: Enterprise Data Lakehouse Build. Ingest 5 years of retail data into ADLS Gen2 using Delta Lake format. Use Microsoft Purview to catalog and classify the data. Create Synapse Serverless SQL Pool Views to serve executive dashboards logically without duplicating data into a dedicated warehouse.

3 Milestone 3: Batch Processing & ETL Orchestration (Month 3)

Objective: Master Azure Databricks, Azure Data Factory (ADF), and Data Migration.

Week 9: Azure Databricks (Managed Spark)

- **Monday:** Databricks Workspace, Clusters (All-Purpose vs. Job), and DBFS.
- **Tuesday:** Apache Spark internals on Databricks (DataFrames, RDDs, Partitioning).
- **Wednesday:** Databricks Delta Lake and Delta Live Tables.
- **Thursday: Hands-on Project:** Deploy an Azure Databricks workspace. Submit a PySpark notebook that reads data from ADLS Gen2, performs a distributed aggregation, and writes to a Delta table.

- **Friday: Use Case & Architecture:** Design a migration strategy for a 100-node on-premise Hadoop cluster, prioritizing cost reduction and moving the processing engine to Azure Databricks.

Week 10: Visual ETL (Azure Data Factory)

- **Monday:** ADF Concepts: Linked Services, Datasets, Pipelines, and Activities.
- **Tuesday:** Mapping Data Flows (Visual, code-free data transformations backed by Spark).
- **Wednesday:** Integration Runtimes (Azure, Self-Hosted, Azure-SSIS).
- **Thursday: Hands-on Project:** Build a visual pipeline in ADF using Mapping Data Flows that joins two datasets from ADLS Gen2, drops nulls, transforms timestamps, and loads the clean data into Synapse.
- **Friday: Use Case & Architecture:** Evaluate Mapping Data Flows in ADF vs. Databricks Notebooks for a team of business analysts who need to build daily reporting pipelines without writing code.

Week 11: Change Data Capture (CDC) & Migration

- **Monday:** Azure Database Migration Service (DMS) architecture.
- **Tuesday:** CDC features in Azure SQL and Cosmos DB Change Feed.
- **Wednesday:** Event-driven data pipelines using Azure Event Grid.
- **Thursday: Hands-on Project:** Configure an Azure DMS task with CDC enabled to stream ongoing database changes from an Azure SQL Database to an ADLS Gen2 Data Lake.
- **Friday: Use Case & Architecture:** Architect a near-real-time synchronization pipeline from an on-premise Oracle database to Azure Synapse Analytics for live BI reporting.

Week 12: Pipeline Orchestration (ADF & Apache Airflow)

- **Monday:** Advanced ADF Control Flow: ForEach, If Condition, Until, Execute Pipeline.
- **Tuesday:** Managed Airflow in Azure Data Factory (formerly Azure Workflow parameters).
- **Wednesday:** Azure Logic Apps for business process automation and notifications.
- **Thursday: Hands-on Project:** Create an ADF pipeline that: 1. Waits for an Event Grid trigger. 2. Triggers a Databricks Notebook. 3. Calls a Logic App to send a Teams message on failure.
- **Friday: Use Case & Architecture:** Design a resilient, self-healing orchestration layer for a daily banking reconciliation process spanning 15 dependent systems using ADF and Logic Apps.

CAPSTONE PROJECT - Milestone 3: Automated Batch ELT Pipeline. Use Azure Event Grid to trigger an ADF pipeline upon file upload to ADLS Gen2. The pipeline triggers a Databricks notebook to perform complex data quality checks and ML feature engineering, and uses the ADF Copy Activity to bulk load the output into a Synapse Dedicated SQL Pool.

4 Milestone 4: Streaming & Real-Time Analytics (Month 4)

Objective: Conquer unbounded data, Event Hubs, Stream Analytics, and Power BI.

Week 13: Message Queuing & Event Ingestion

- **Monday:** Decoupling with Azure Service Bus (Queues/Topics).
- **Tuesday:** Azure Event Hubs: Partitions, Consumer Groups, and Throughput Units.
- **Wednesday:** Event Hubs Capture (automatic delivery to ADLS Gen2) and Kafka compatibility.
- **Thursday: Hands-on Project:** Create an Azure Event Hub. Write a Python producer script to stream mock IoT temperature sensor data, and configure Event Hubs Capture to save batches to ADLS Gen2.
- **Friday: Use Case & Architecture:** Architect a global telemetry ingestion point for 5 million mobile games, ensuring high throughput, partitioning by user ID, and low-latency processing.

Week 14: Real-Time Processing (Azure Stream Analytics)

- **Monday:** Azure Stream Analytics (ASA) architecture and SQL-like query language.
- **Tuesday:** Windowing functions: Tumbling, Hopping, Sliding, and Session windows.
- **Wednesday:** Handling late-arriving data and out-of-order events in ASA.
- **Thursday: Hands-on Project:** Write an ASA job that reads from Event Hubs, groups user clicks into 5-minute Tumbling Windows, counts the clicks, and writes the output to Azure SQL Database.
- **Friday: Use Case & Architecture:** Troubleshoot a scenario where a streaming dashboard is showing inaccurate daily aggregations due to mobile devices going offline and uploading data late.

Week 15: Advanced Streaming (Spark Structured Streaming)

- **Monday:** Apache Spark Structured Streaming on Azure Databricks.
- **Tuesday:** Micro-batch processing vs. Continuous Processing.
- **Wednesday:** Integrating Event Hubs with Databricks using the Kafka API.
- **Thursday: Hands-on Project:** Write a Databricks notebook using Spark Structured Streaming to consume data from Event Hubs, aggregate by window, and append the results into a Delta Lake table.
- **Friday: Use Case & Architecture:** Design a pipeline to anonymize incoming healthcare HL7 messages in real-time using Databricks before streaming them into a Synapse workspace for analysis.

Week 16: Business Intelligence & Power BI

- **Monday:** Power BI Architecture: Desktop, Service, Datasets, and Dataflows.
- **Tuesday:** DirectQuery vs. Import mode vs. Dual mode.
- **Wednesday:** Row-Level Security (RLS) and DAX basics.
- **Thursday:** **Hands-on Project:** Connect Power BI Desktop to your Synapse Dedicated Pool using DirectQuery. Create a real-time dashboard monitoring sales, apply a filter, and set up an RLS role.
- **Friday:** **Use Case & Architecture:** Design the BI access layer for an enterprise where 500 analysts need to query centralized data, ensuring regional sales managers can only see data for their specific regions (RLS).

CAPSTONE PROJECT - Milestone 4: Real-Time Fraud Detection. Build a Python script simulating credit card swipes sent to Azure Event Hubs. Create an Azure Stream Analytics job that uses a Sliding Window to detect if a card is used in two different cities within 10 minutes, flags it, routes it to an Azure SQL Database, and displays the live fraud alerts in a streaming Power BI dashboard.

5 Milestone 5: Machine Learning Operationalization (Month 5)

Objective: Embed AI into data pipelines and manage the MLOps lifecycle with Azure ML.

Week 17: In-Database Machine Learning (Synapse PREDICT)

- **Monday:** Synapse Analytics Machine Learning integration.
- **Tuesday:** T-SQL PREDICT function to score data natively in Dedicated SQL Pools.
- **Wednesday:** Training models using Spark in Synapse and deploying to SQL Pools.
- **Thursday:** **Hands-on Project:** Use Spark in Synapse to train a logistic regression model on customer data to predict churn. Export the model, and run a T-SQL PREDICT query to score live data.
- **Friday:** **Use Case & Architecture:** Architect a system to automatically forecast daily inventory needs for 1,000 retail stores using purely SQL-based pipelines to minimize data movement.

Week 18: Azure Machine Learning (AML) Foundations

- **Monday:** Azure Machine Learning Studio, Compute Instances, and Compute Clusters.
- **Tuesday:** Automated ML (AutoML) vs. Designer (Drag-and-drop) vs. Custom Notebooks.
- **Wednesday:** AML Managed Endpoints: Real-time and Batch Inference.
- **Thursday:** **Hands-on Project:** Use Azure AutoML to train a classification model on a cleaned ADLS dataset. Deploy the best model to a Managed Online Endpoint and send a test REST payload.

- **Friday: Use Case & Architecture:** Design a workflow for data scientists to train custom PyTorch models on GPU compute clusters while securely accessing training data stored in a governed ADLS data lake.

Week 19: MLOps & Azure ML Pipelines

- **Monday:** Azure ML Managed Feature Store concepts.
- **Tuesday:** Azure ML Pipelines (Components and Environments) for MLOps orchestration.
- **Wednesday:** Data Drift detection and Model Monitoring in AML.
- **Thursday: Hands-on Project:** Build an Azure ML Pipeline (Preprocess → Train → Evaluate → Register). Execute the pipeline using the Python SDK from a Notebook.
- **Friday: Use Case & Architecture:** Architect an MLOps lifecycle that detects data drift in a production recommendation engine and automatically triggers a retraining pipeline using Azure Monitor/Event Grid.

Week 20: Unstructured Data & Azure Cognitive Services

- **Monday:** Azure Computer Vision and Azure Form Recognizer (Document Intelligence).
- **Tuesday:** Azure Cognitive Service for Language (NLP) and Translator.
- **Wednesday:** Integrating Cognitive Services within Azure Data Factory pipelines.
- **Thursday: Hands-on Project:** Write an Azure Function triggered by an ADLS Gen2 upload that sends a scanned PDF invoice to Azure Form Recognizer, extracts key-value pairs, and saves them as JSON.
- **Friday: Use Case & Architecture:** Design a scalable pipeline to process terabytes of uploaded user images, extract text (OCR), tag objects, and make the metadata searchable using Azure Cognitive Search.

CAPSTONE PROJECT - Milestone 5: End-to-End MLOps Pipeline. Use Azure Data Factory to orchestrate a pipeline that extracts features via Synapse, pushes them to Azure ML, triggers an Azure ML Pipeline to train a custom model, registers the model in the AML Model Registry, and sets up a batch endpoint to score new data nightly.

6 Milestone 6: Security, Governance & Exam Prep (Month 6)

Objective: Secure the perimeter, implement Data Mesh, and prepare for DP-203 certification.

Week 21: Security, Azure AD & Networking

- **Monday:** Microsoft Entra ID (Azure AD), Service Principals, and Managed Identities.
- **Tuesday:** VNet Integration: Private Endpoints, Service Endpoints, and Azure Bastion.
- **Wednesday:** Azure Key Vault for secrets and Customer-Managed Keys (CMK).
- **Thursday: Hands-on Project:** Create a Private Endpoint for an ADLS Gen2 account. Disable public access. Use a VM within the same VNet to securely access the storage account via private IP.
- **Friday: Use Case & Architecture:** Design a highly secure data perimeter for a healthcare provider ensuring no data exfiltration can occur, utilizing VNet Data Exfiltration protection in Synapse.

Week 22: Advanced Data Governance (Purview)

- **Monday:** Microsoft Purview: Data map, classifications, and sensitivity labels.
- **Tuesday:** Data policies and access provisioning via Purview.
- **Wednesday:** Integrating Purview with Azure Synapse and Databricks for holistic lineage.
- **Thursday: Hands-on Project:** Define a custom classification rule in Microsoft Purview for a proprietary employee ID format. Scan a data lake to automatically discover and tag this data.
- **Friday: Use Case & Architecture:** Architect a Data Mesh for a global conglomerate with 5 independent domains, ensuring centralized governance via Microsoft Purview but decentralized data engineering.

Week 23: Compliance & Data Privacy

- **Monday:** Dynamic Data Masking (DDM) in Azure SQL and Synapse.
- **Tuesday:** Always Encrypted and Transparent Data Encryption (TDE).
- **Wednesday:** Azure Policy and Microsoft Defender for Cloud (Auditing and Alerts).
- **Thursday: Hands-on Project:** Implement Dynamic Data Masking on an Azure SQL Database table containing PII. Create a read-only user and verify that they see masked data (e.g., xxx-xxxx) for emails.
- **Friday: Use Case & Architecture:** Design a pipeline that ingests third-party customer data, automatically identifies PII, tokenizes it using Azure Key Vault secrets, and stores the safe version in Synapse.

Week 24: DP-203 Certification Preparation

- **Monday:** Review DP-203 (Data Engineering on Microsoft Azure) Exam Objectives.
- **Tuesday:** Whitepaper/Docs Review: Data architecture guide for Azure.
- **Wednesday:** Full-length Mock Exam 1 (Identify weak domains).
- **Thursday:** Targeted review of weak domains (e.g., Databricks vs. Synapse Spark, ASA windows).
- **Friday:** Full-length Mock Exam 2.

CAPSTONE PROJECT - Milestone 6: Secure, Governed Data Mesh. Build a multi-subscription architecture. Subscription A ingests streaming PII data and uses Azure Functions to tokenize it. Subscription B (Analytics) hosts Synapse. Subscription C (Governance) uses Microsoft Purview and Private Endpoints to lock down all traffic, ensuring a zero-trust network boundary.

7 6-Month Progress Tracking Timetable

Week	Primary Focus	Key Project / Capstone	Status
1	ADLS Gen2 & Storage	Storage Lifecycle & Hierarchical Namespace	☐
2	Azure SQL & PostgreSQL	Azure SQL Serverless Failover Simulation	☐
3	Azure Cosmos DB	Cosmos DB Partitioning & Global Writes	☐
4	Hybrid Ingestion & Caches	Cap 1: E-Commerce Multi-Tier Data	☐
5	Synapse Analytics Basics	COPY into Synapse Dedicated SQL Pool	☐
6	Synapse Optimization	Distribution Types Benchmark	☐
7	Synapse Serverless SQL	Querying ADLS via CETAS & Logical Views	☐
8	Microsoft Purview	Cap 2: Enterprise Data Lakehouse	☐
9	Azure Databricks	PySpark Aggregation on ADLS Gen2	☐
10	Azure Data Factory (ADF)	ADF Mapping Data Flows Visual ETL	☐
11	Azure DMS & CDC	Azure SQL to ADLS CDC Pipeline	☐
12	ADF Orchestration	Cap 3: Automated Batch ELT	☐
13	Azure Event Hubs	IoT Producer/Consumer Scripting	☐
14	Azure Stream Analytics	ASA Tumbling Windows to Azure SQL	☐
15	Spark Structured Streaming	Databricks Streaming from Event Hubs	☐
16	Power BI	Cap 4: Real-Time Fraud Telemetry	☐
17	Synapse PREDICT	Churn Prediction T-SQL Models	☐
18	Azure ML Foundations	AutoML Deployment & REST Endpoint Test	☐
19	Azure ML Pipelines	AML Pipeline Execution via Notebook	☐
20	Cognitive Services (AI APIs)	Cap 5: End-to-End MLOps Pipeline	☐
21	Entra ID & Private Endpoints	ADLS Private Endpoint implementation	☐
22	Purview Governance	Custom Classifications & Scans	☐
23	Dynamic Data Masking	Azure SQL DDM on PII Columns	☐
24	Exam Prep (DP-203)	Cap 6: Secure Data Mesh Design	☐

"Consistency is the architecture of mastery."

Best of luck on your 6-month journey to becoming a Master Azure Data Engineer!